Roll No. ☐☐☐☐☐☐☐☐☐☐☐ Total No. of Pages : 02

Total No. of Questions : 13

B.Pharma (Sem.-6)

BIOPHARMACEUTICS AND PHARMACOKINETICS (THEORY)

Subject Code : BP-604T

M.Code : 77989

Date of Examination : 16-05-2024

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. **Explain briefly :**

   a) Name plasma proteins responsible for drug - protein binding.

   b) How is elimination half-life of a drug calculated from slope of elimination phase?

   c) What is renal clearance and how is it calculated?

   d) What is meant by very high Vd?

   e) Mention four reasons for reduced oral bioavailability of drugs.

   f) Mention the non-renal routes of drug elimination.

   g) Draw the plasma - time curve for oral administration of drug for one compartment open model kinetics.

   h) What is pinocytosis?

   i) What is total clearance and how it is calculated?

   j) What is meant by biliary excretion?

SECTION - B

2. What is meant by non-linear pharmacokinetics. Discuss the factors responsible for non-linear pharmacokinetics of drugs.

3. Discuss the different absorption processes and mention their limitations.

4. IV injection of a drug to a 70Kg patient gave drug serum concentration of 1.2 microgm / ml and 0.3 microgm / ml at 2 hr and 5 hr, respectively. Calculate T1/2 of the drug assuming first order kinetics.

SECTION-C

5. If the T1/2 of a drug is 12 hr, how long would it take for 150 mg dose to get 30% eliminated?

6. Differentiate between one and two compartment open models. Draw typical plasma-time profiles for them after oral drug administration and label various phases in the profiles.

7. Discuss the Wagner Nelson method with suitable equations.

8. Comment on the methods employed for enhancing the dissolution rate of poorly soluble drugs.

9. Considering a IV bolus dose and simultaneous IV infusion, prove that for instantaneously achieving steady state a loading dose of R/K (infusion rate/elimination half-life) is necessary.

10 Differentiate between relative and absolute bioavailability.

11. Enumerate pharmaceutical factors influencing drug absorption and discuss any one in detail

12. What is Sigma-Minus method? Explain the method of calculating elimination rate constant by this method with the help of suitable equations.

13. Comment on factors affecting drug distribution in the body.

NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.

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